Evaluation and Monoculture in Large Language Models

1 Introduction

Monoculture is a big deal in a lot of field, where different players in a market act similarly. This is a bad thing because of the loss of diversity, increase in monopoly, etc. In AI, this would mean that frontier models are converging not just in overall quality but also capabilities (which domains models are strong and weak at). Whether this is good or bad has no single answer, and we don't think it should: differentiated frontier models serve users who need a specific strength, while convergence on one excellent, broadly capable default can serve users who just want the best available option. Both are real outcomes worth telling apart — not just whether they occur, but why.

Right now AI development occurs through evaluation platforms; the process contributing to convergence are observable. This allows us to ask a question little studied in the monoculture literature; what produces the monoculture, looking specifically at how evaluation impacts monoculture. This is not a hypothetical candidate: firms are documented to optimize specifically toward what shared evaluations measure, which makes evaluation an active participant in whatever convergence results, not a passive bystander to it. Evaluation is also distinctive because we can control them, and so we can actually control this lever. We model the strategic game firms play under a centralized evaluation, and find equilibrium behavior is governed by two related but distinct forces: whether a stable equilibrium exists at all, and, conditional on one existing, whether it is differentiated or homogeneous. That characterization becomes a design tool: which features of an evaluation protocol push firms together, and which let them stay apart — giving whoever has a stake in the outcome a precise lever to reason with, rather than a general intuition.

Our central question is to understand the role that evaluation plays in shaping heterogeneity in monoculture. We characterize its impacts through a simple yet rich model, and use this to give recommendations on how changing the way centralized evaluations are structured can lead to more or less heterogeneity. We incorporate factors such as social welfare, and reinforce this with an empirical investigation of how our predictions correspond to practice.

2 Model and Preliminaries

2.1 Model Basics

Our core model consists of N firms, each of which competes on K dimensions. Firm i invests amount $x_{i,k}$ in category k, resulting in an output $\theta_i = F_i(\mathbf{x}_i) : \mathbb{R}_+^K \to \mathbb{R}_+^K$. Model developers additionally suffer a cost $c(\sum_i x_{i,k})$, for some cost function c.

The goal is to maximize performance according to a central evaluator, which ranks models based on "win-rate"; performance against other models. Here, battles are against other models on the K different topics, each with probability w_k, where $\mathbf{w} \in \Delta^{K-1}$ is a vector set by the central evaluator based on submitted prompts. That is, firm i selects $\mathbf{x}_i$ to maximize the win-rate against other models:

$$\max_{\mathbf{x}_i} \frac{1}{N} \sum_{i' \neq i} \mathbb{E}_{y \sim \text{Cat}(\mathbf{w})}[\sigma(F_i(\mathbf{x}_i)_y - F_i(\mathbf{x}_{i'})_y)] - c(\sum x_{i,k}).$$

Here, σ is some function that captures win-probability, with the requirements that $\sigma(0) = 0.5$, and that σ is non-decreasing, differentiable, and bounded by 0-1, with $\sigma(x) = 1 - \sigma(-x)$. All of these properties are satisfied by the

Bradley-Terry preference model (among other such models), and captures essentially any reasonable preference model. In essence this optimization problem corresponds to firms optimizing for some linear function of win-rate. For example, if firms were ranked based on overall win-rate, or some aggregation of per-category Bradley-Terry coefficients, then this model would perfectly capture that. What this model *does not* capture is the scenario where models are ranked according to one Bradley-Terry coefficient-per model; such a ranking is misspecified due to the per-category nature of productions (and because a weighted average of logistic functions is itself not a logistic function).

Our model can be viewed as an extension of Chen et al. (2026). In their setting, they consider a function $v(\theta_i, e_i)$ (for v bounded, monotonic, and concave), where θ_i is the inherent ability of the model, and firms strategize over their selection of e_i. Our production function $F_i(\mathbf{x}_i)$ is a generalization of this to multiple dimensions, and we can recover the original setting by letting $K = 1$ and $f_i(x) = v(\theta_i, x)$.

Our primary difference from prior work (Chen et al., 2026) is our modeling of the win-rates. In prior work, they view rewards for each rank, $R_1, \ldots, R_N$, which induces discontinuity as no reward is achieved for straddling between R_i and R_{i-1} (that is, it is better to be ϵ better than the rank below you. In our setting, we view rewards due to some win-rate function σ, which smoothes this out. We later show that even a ranking version (where a firm gets a reward based on their rank) with smoothing can eliminate discontinuities while achieving generality. Our production function and cost combination (F_i and c) is more general than prior work ($v(\theta_i, \cdot)$ and c). We additionally note that our selection of (F_i, c) is equivalent to the (F_i, c_i) case; where costs are individualized per-firm:

Proposition 1 (Reduction to a common cost function)**.** *Fix any firm i with production function F_i and cost function c_i, where $c_i : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ is a strictly increasing bijection with $c_i(0) = 0$. Let c be any other such cost function. Then there exists a production function $\tilde{F}_i$ such that the optimization problem defined by (F_i, c_i) and the optimization problem defined by $(\tilde{F}_i, c)$ have identical solutions: for every action vector $\mathbf{x}_i \in \mathbb{R}^K_{\geq 0}$ there is a corresponding vector $\mathbf{z}_i \in \mathbb{R}^K_{\geq 0}$ with*

$$c_i\big(\|\mathbf{x}_i\|_1\big) = c\big(\|\mathbf{z}_i\|_1\big) \qquad \text{and} \qquad F_i(\mathbf{x}_i) = \tilde{F}_i(\mathbf{z}_i),$$

and this correspondence $\mathbf{x}_i \leftrightarrow \mathbf{z}_i$ is a bijection.

Proof. Define the scalar map

$$g_i(T) := c_i^{-1}\big(c(T)\big), \qquad T \geq 0,$$

so that, by construction,

$$c_i\big(g_i(T)\big) = c(T) \qquad \text{for all } T \geq 0. \tag{1}$$

Since c_i and c are strictly increasing bijections of $\mathbb{R}_{\geq 0}$, so is g_i. In particular g_i^{-1} exists and is itself a strictly increasing bijection of $\mathbb{R}_{\geq 0}$.

Given $\mathbf{x}_i \in \mathbb{R}^K_{\geq 0}$ with total effort $T_i^x := \|\mathbf{x}_i\|_1$, define the *reference-unit* vector

$$\mathbf{z}_i := \frac{g_i^{-1}\big(T_i^x\big)}{T_i^x}\,\mathbf{x}_i,$$

i.e. $\mathbf{x}_i$ rescaled, keeping its direction fixed, so that its new total is $\|\mathbf{z}_i\|_1 = g_i^{-1}(T_i^x)$. Applying g_i to both sides of this last identity and using $g_i\big(g_i^{-1}(T)\big) = T$ gives

$$g_i\big(\|\mathbf{z}_i\|_1\big) = T_i^x. \tag{$*$}$$

Combining $(*)$ with (1) evaluated at $T = \|\mathbf{z}_i\|_1$:

$$c\big(\|\mathbf{z}_i\|_1\big) = c_i\Big(g_i\big(\|\mathbf{z}_i\|_1\big)\Big) = c_i(T_i^x) = c_i\big(\|\mathbf{x}_i\|_1\big),$$

so cost is exactly preserved.

Define, for $\mathbf{z} \in \mathbb{R}^K_{\geq 0}$,

$$\tilde{F}_i(\mathbf{z}) := F_i\left(\frac{g_i\big(\|\mathbf{z}\|_1\big)}{\|\mathbf{z}\|_1}\,\mathbf{z}\right).$$

Evaluating at $\mathbf{z} = \mathbf{z}_i$ and using $(*)$:

$$\tilde{F}_i(\mathbf{z}_i) = F_i\left(\frac{g_i(\|\mathbf{z}_i\|_1)}{\|\mathbf{z}_i\|_1}\,\mathbf{z}_i\right) = F_i\left(\frac{T_i^x}{\|\mathbf{z}_i\|_1}\cdot\frac{g_i^{-1}(T_i^x)}{T_i^x}\,\mathbf{x}_i\right) = F_i\left(\frac{T_i^x}{\|\mathbf{z}_i\|_1}\cdot\frac{\|\mathbf{z}_i\|_1}{T_i^x}\,\mathbf{x}_i\right) = F_i(\mathbf{x}_i),$$

using $\|\mathbf{z}_i\|_1 = g_i^{-1}(T_i^x)$ in the middle step. Thus $F_i(\mathbf{x}_i) = \tilde{F}_i(\mathbf{z}_i)$ as required. Since g_i and g_i^{-1} are both strictly increasing bijections of $\mathbb{R}_{\geq 0}$, the map $\mathbf{x}_i \mapsto \mathbf{z}_i$ above is itself a bijection of $\mathbb{R}^K_{\geq 0}$, with inverse $\mathbf{z}_i \mapsto \mathbf{x}_i = \frac{g_i(\|\mathbf{z}_i\|_1)}{\|\mathbf{z}_i\|_1}\mathbf{z}_i$. □

We then define the heterogeneity of allocation between a pair of firms as $H_{i,i'} = \cos(\theta_i, \theta_{i'}) = \frac{\theta_i \cdot \theta_{i'}}{\|\theta_i\|\|\theta_{i'}\|}$. We then define the aggregate heterogeneity as $H = \frac{1}{\binom{N}{2}}\sum_{i,i'} H_{i,i'}$. Our definition of H aligns with natural intuitions from the algorithmic monoculture literature, which measures similarity in outputs between different firms.

3 Equilibrium and Heterogeneity

3.1 Equilibrium with Two Firms

We first tackle the problem with $K = 1$, or one category, so that vector-valued F_i reduces to scalar valued (but heterogeneous) f_i. Suppose for simplicity that $N = 2$, then the optimization problem is

$$U_1(x_1; x_2) = \sigma(f_1(x_1) - f_2(x_2)) - c(x_1), \qquad U_2(x_2; x_1) = \sigma(f_2(x_2) - f_1(x_1)) - c(x_2).$$

We note that for both U_1 and U_2, due to convexity and unboundedness of c (and that $U_i(0) \geq 0$, we have that there exists a maximizer for each. If U_1 and U_2 are concave, then they have unique maximizers, and so for any x_2, there exists a unique maximizer x_1. Next, let $h(x_1)$ be the composition of best-response functions; $h(x_1) = \mathrm{BR}_1(\mathrm{BR}_2(x_1)) - x_1$. Note that $h(0) \geq 0$, and for X large, $h(X) < 0$. By intermediate value theorem, there exists an x_1, x_2, so $h(x) = 0$ and x_1 and x_2 are best responses to each other.

We next determine whether that critical point is a maximizer by investigating concavity. Let $z = f_1(x_1) - f_2(x_2)$, then take second derivatives to identify concavity:

$$U_1''(x_1; x_2) = f_1''(x_1)\sigma'(z) + f_1'(x_1)^2\sigma''(z) - c''(x_1).$$

To achieve concavity we need that

$$f_1''(x_1)\sigma'(z) + f_1'(x_1)^2\sigma''(z) - c''(x_1) \leq 0.$$

A sufficient condition for this is that

$$f_1''(x_1)\sigma'(z) + f_1'(x_1)^2\sigma''(z) \leq 0,$$

or that

$$f_1'(x_1)^2\frac{\sigma''(z)}{\sigma'(z)} \leq -f_1''(x_1).$$

Finally rearranging, and taking the supremum over z gives

$$\sup_z \frac{\sigma''(z)}{\sigma'(z)} \leq -\frac{f_1''(x_1)}{f_1'(x_1)^2}\forall x_1.$$

Therefore, critical points are maximizer, and mutual critical points are Nash Equilibria.

Broadly, the key idea is that the curvature of f dictates whether an equilibrium exists. If f increases too quickly (e.g., linear), then no equilibrium exists because the utility function becomes non-concave, and best responses can ping-pong between $x_1 = 0$ and x_1 is large. On the flipside, for curved functions, returns diminish quickly, allowing for concavity in the utility function, and the presence of equilibrium. Moreover, the role that σ'' plays is similar; if $\sigma''(z) \gg \sigma'(z)$, then the win-rate is essentially a step-function, which leads to firms incentivizing to barely achieve victory over other firms. The key innovation that allows us to overcome this is the (relative) smoothness of σ.

3.2 Existence of General Equilibria

We can take the ideas from before to prove a general theorem for when equilibria exist across values of N and K:

Theorem 2 (Equilibrium Existence)**.** *Let F_i be a prodduction function and c the cost. Then there exists an equilibrium to the game if*

$$-\nabla^2 g_{i,k}(x) \succeq \sup_z \frac{\sigma''(z)}{\sigma'(z)} \cdot \nabla g_{i,k}(x) \nabla g_{i,k}(x)^\top \qquad \forall x,\ \forall i,k.$$

Corollary 3. *There exists a unique equilibrium if*

$$-\nabla^2 g_{i,k}(x) \succ \sup_z \frac{\sigma''(z)}{\sigma'(z)} \cdot \nabla g_{i,k}(x) \nabla g_{i,k}(x)^\top \qquad \forall x,\ \forall i,k.$$

Proof. This follows by noting that a strictly convex function has at one critical point, and so the maximizer is unique; that is, best responses are all unique. □

We note that the requirements here are almost identical to that of the $N = 2$, $K = 1$ scenario, with just the terms themselves translated to higher dimensions.

3.3 Characterizing Homogeneity

We next show that for general N and K, the zero-heterogeneity condition holds when $F_1 = F_2 = \cdots = F_N$, assuming that an equilibrium exists:

Lemma 4. *Suppose F_i is constant. If an equilibrium exists, according to Theorem 1, at least one equilibrium produces $H_{i,i'} = 0$ for all i and i'.*

Proof. Suppose $F_1 = F_2 \cdots F_N$, and suppose that Theorem 1 holds strictly (that is, a unique equilibrium holds). Then there exists an equilibrium. We note that permuting the indices still maintains an equilibrium, and so the two equilibrium must be the same. Therefore, all firms play the same strategy, and so there must be a symmetric equilibrium. This results in $H = 0$. □

Proof. □

3.4 Identifiability of Monoculture

We note that F_i constant is only one way to produce monoculture. Rather, in general, $H = 0$ implies nothing about the underlying F_i. As an example, consider the case when $N = 2, K = 2$ and $F_i = A_i \log(x + 1)$ with $w = [0.4, 0.6]$. Fix A_1 as

$$A_1 = \begin{pmatrix} 1.0 & 0.2 \\ 0.3 & 0.8 \end{pmatrix},$$

then any A_2 satisfying

$$A_2(a,\beta) = \begin{pmatrix} a & \dfrac{y_1^* - q_1(\beta)\,a}{q_2(\beta)} \\ \dfrac{\beta_1 - w_1 a}{w_2} & \dfrac{y_2^* - q_1(\beta)\left(\dfrac{\beta_1 - w_1 a}{w_2}\right)}{q_2(\beta)} \end{pmatrix}$$

where

$$q(\beta) = \log\!\left(\frac{\beta}{4\,t^*(\beta_1+\beta_2)}\right), \qquad t^*(s) = -1 + \tfrac{1}{2}\sqrt{4+s}$$

$$C = \Big\{\beta \in \mathbb{R}^2 : w^\top y^* = \beta \cdot q(\beta)\Big\}, \qquad y^* = A_1\, q(A_1^\top w),$$

satisfies $H = 0$. We note that $H = 0$ occurs much more commonly for solutions with $F_1 \neq F_2$; and note that this is a general scenario rather than a specific scenario.

Theorem 5 (Homogeneity without Equality)**.** *Under (H1)–(H2) and (C), the set*

$$\mathcal{T} := \{\boldsymbol{\phi} \in U : H_{i,i'}(\boldsymbol{\phi}) = 0 \;\; \forall\, i, i'\}$$

is a smooth submanifold of U of dimension

$$\dim \mathcal{T} = Nk - (K-1)(N-1),$$

while the identity locus

$$\mathcal{I} := \{\boldsymbol{\phi} \in U : \phi_1 = \cdots = \phi_N\}$$

has dimension $\dim \mathcal{I} = k$ and satisfies $\mathcal{I} \subseteq \mathcal{T}$ by Lemma 1. Consequently,

$$\operatorname{codim}(\mathcal{I} \text{ in } \mathcal{T}) = (N-1)(k-K+1),$$

which is strictly positive whenever $k \geq K$. In this regime $\mathcal{I}$ has Lebesgue measure zero within $\mathcal{T}$: a generic point of $\mathcal{T}$ satisfies $H_{i,i'} = 0$ for every pair while $F_{\phi_i} \neq F_{\phi_{i'}}$ for some pair i, i'.

Corollary 6 (Verification for log-linear production)**.** *Let $F_\phi(x) = A\log(1+x)$ with $\phi = \operatorname{vec}(A) \in \mathbb{R}^{K^2}$, so $k = K^2$. Then $D_\phi F_\phi(x) \cdot dA = dA\,\log(1+x)$ for any perturbation dA of the matrix A. For any target $v \in \mathbb{R}^K$, the choice $dA = v\,\log(1+x)^\top / \|\log(1+x)\|^2$ satisfies $dA\,\log(1+x) = v$, so $D_\phi F_\phi(x)$ is surjective onto $\mathbb{R}^K$ whenever $\log(1+x) \neq 0$, i.e., whenever $x \neq 0$. Since equilibrium investment is strictly positive in at least one category at any interior equilibrium, condition (C) holds at every equilibrium point for the log-linear family, and the theorem applies with $k = K^2$. For $N = K = 2$, this gives $k = 4$ and*

$$\operatorname{codim}(\mathcal{I} \text{ in } \mathcal{T}) = (2-1)(4-2+1) = 3,$$

matching the explicit three-parameter family of tying matrices $A_2(a,\beta,\gamma)$ obtained directly from the direction-matching equation $A_2\, q(A_2^\top w) = \lambda\, y^*$.

The implication from this is homogenization does not require identical production functions; rather even wildly diverging technologies can lead to similar outputs. That is, we cannot bound homogenization based on similariy in the production function;

$$A_2 \approx \begin{pmatrix} 0.0025 & 1.4142 \\ 0.9497 & 0.0082 \end{pmatrix}$$

with the resulting equilibrium as

$$x_1^* = (0.0855,\ 0.0481), \qquad x_2^* = (0.0671,\ 0.0667).$$

Here, even though A_1 and A_2 are quite different, they can result in what appears to be homogeneity.

Proposition 7. *The win function σ does not impact which F_1 and F_2 cause homogenization.*

Proof. It's always $\sigma'(0)$ and so it might impact the constant, but otherwise it's non-impacted. As long as existence and uniqueness are satisifed, you're good. As long as σ satisfies the bound on $\frac{\sigma''}{\sigma'}$. □

We finally note that if homogeneity ($F_i(\mathbf{x}_i)$ being constant) holds for a given c, then small changes in c can break heterogeneity. That is, if costs change over time (e.g., diminishing costs for developing models), then with sufficiently large timesteps, only F_i constant can guarantee homogeneity ($H = 0$).

3.5 When can you identify Monoculture?

To prove this, we first note that similarity in F_i can guarantee a small H:

Lemma 8 (Similarity in Production Functions)**.** *Suppose the following hold. First* $\sup_x \|F_i(\mathbf{x}) - F_{i'}(\mathbf{x})\| \leq \epsilon$. *Second, the gradient* $\Phi(\mathbf{x}) = \langle \nabla F_1, \nabla F_2, \ldots, \nabla F_N \rangle$ *is strongly monotonic by* μ*;*

$$(\Phi(\mathbf{x}) - \Phi(\mathbf{x}')) \cdot (\mathbf{x} - \mathbf{x}') \leq -\mu \|\mathbf{x} - \mathbf{x}'\|^2.$$

Third, F is L-Lipschitz, and $\|\nabla F_i(\mathbf{x}) - \nabla F_j(\mathbf{x})\|_\infty \leq \epsilon$. *Then*

$$\|F_i(\mathbf{x}_i^*) - F_j(x_j^*)\|_\infty \leq \epsilon(1 + L/\mu),$$

and

$$H_{i,j} \geq 1 - 2\epsilon(1 + C/\mu)/m,$$

where m is a lower bound on $\|\theta_i\|$.

Fix a pair of firms i, i' and let $\{c_\gamma\}_{\gamma \in \Gamma}$, $\Gamma \subseteq \mathbb{R}$ open, be a smooth one-parameter family of admissible costs (e.g. $c_\gamma(t) = \gamma \cdot c(t)$), and let $\mathcal{W} \subseteq \Delta^{K-1}$ be an open neighborhood of some w_0. Write $x_i^*(\gamma, w)$ for firm i's equilibrium investment under (c_γ, w), smooth on $\Gamma \times \mathcal{W}$ by (H1).

(D1) (Richness.) At some (γ_0, w_0) with equilibrium $x_0 := x_i^*(\gamma_0, w_0)$, the Jacobian $D_{(\gamma,w)}\, x_i^*(\gamma_0, w_0)$, a $K \times K$ matrix, is invertible.

(D2) (Coincidence-forcing.) No discrepancy between $x_i^*(\gamma, w)$ and $x_{i'}^*(\gamma, w)$ survives every perturbation in $\Gamma \times \mathcal{W}$: if $x_i^*(\gamma_0, w_0) \neq x_{i'}^*(\gamma_0, w_0)$, some (γ, w) arbitrarily close to (γ_0, w_0) breaks the tie $H_{i,i'} = 0$.

- Both $F_i, F_{i'}$ are real-analytic on their domain.

Theorem 9 (Joint cost–weight robustness identifies technology up to a pointwise scale)**.** *Under (H1), (D1), (D2), suppose $H_{i,i'}(\gamma, w) = 0$ for all $(\gamma, w) \in \Gamma \times \mathcal{W}$. Then $F_i(x) \parallel F_{i'}(x)$ for every x in the (connected) domain of $F_i, F_{i'}$, i.e., $F_i(x) = \lambda(x)\, F_{i'}(x)$ for some scalar function $\lambda(\cdot) > 0$.*

Proof. By (D1) and the inverse function theorem, $(\gamma, w) \mapsto x_i^*(\gamma, w)$ is a diffeomorphism from a neighborhood of (γ_0, w_0) onto an open neighborhood $U \subseteq \mathbb{R}^K$ of x_0. By (D2), $x_i^*(\gamma, w) = x_{i'}^*(\gamma, w) =: x(\gamma, w)$ throughout this neighborhood, since any discrepancy would generically be detected and would break the tie hypothesis, contradicting $H_{i,i'}(\gamma, w) = 0$ for every (γ, w) in the family. Because x_i^* alone is already a diffeomorphism onto U, the shared equilibrium point $x(\gamma, w)$ sweeps out the entire open set U, not merely a lower-dimensional curve within it. The tie hypothesis $F_i(x_i^*(\gamma, w)) \parallel F_{i'}(x_{i'}^*(\gamma, w))$ therefore applies directly and pointwise, with no chain-rule integration required, giving

$$F_i(x) \parallel F_{i'}(x) \qquad \forall x \in U.$$

Parallel-ness of $F_i(x)$ and $F_{i'}(x)$ is equivalent to the vanishing of every 2×2 minor of the $K \times 2$ matrix $[\, F_i(x) \;\; F_{i'}(x) \,]$. Each such minor is a real-analytic function of x by real-analyticity of $F_i, F_{i'}$, and each vanishes on the nonempty open set U; by the identity theorem for real-analytic functions, each minor therefore vanishes on the entire connected domain. Hence $F_i(x) \parallel F_{i'}(x)$ for every x in the domain, i.e., $F_i(x) = \lambda(x) F_{i'}(x)$ for some scalar function $\lambda(x) > 0$. □

4 How Evaluations Impact Homogeneity

4.1 Smooth Utilities and the Presence of Equilibria

Our analysis thus far has relied on the assumption that firms optimize for win-rates, with rewards essentially linear in win-rate. Here, we show that such an assumption can be partially relaxed, to consider scenarios where we optimize for smoothed-rank functions. That is, functions of the form

$$U_i(x_i; x_{-1}) = \sum_{r=1}^{N-1} \Delta_r \sum_k w_k \sigma(\frac{F_i(\mathbf{x}_i)_k - F_{R_i}(\mathbf{x}_{R_i})_k}{\tau}) - c(\sum_i x_{i,k})$$

Intuitively, this corresponds to scenarios where there is estimation error in the rank (e.g., Bradley-Terry uncertainties), and so exactly splitting ranks R_i and R_{i-1} is stochastic. This smoothening is what allows us to escape the discontinuities from prior work (Chen et al., 2026).

Corollary 10. *If*

$$U_i(x_i; x_{-1}) = \sum_{r=1}^{N-1} \Delta_r \sum_k w_k \sigma(\frac{F_i(\mathbf{x}_i)_k - F_{R_i}(\mathbf{x}_{R_i})_k}{\tau}) - c(\sum_i x_{i,k})$$

then an equilibrium exists under the same conditions as before.

Proof. We can use the same derivation as before, and note that we bound σ' and σ'' over all z; therefore, changes to the details inside σ have no impact. The only change is the constant term going from $\frac{1}{N-1} w_k$ to $w_k \Delta_r$; however, both are positive, and do not impact concavity (due to no changes in the sign), and so the original Theorem holds. □

4.2 Variations in Ranking Functions and Battle Distributions

5 Implications for Evaluation

5.1 Identifying Monoculture and the Implications for Collusion

How do we measure monoculture? Monoculture is inherently a dynamic process. We need many time points because functions rae high dimensional. An approximate version could also be developed depending on the number of points analyzed

Implications for Collusion This takes some ideas from the collusion literature.

5.2 Monoculture in Theory and in Practice

Defining "Style" Differing investment categories have an objective diretion of improvement. However, and inspried by this, at gridlock, firms can also differentiate (for free) on brand/style. In our model, this entails investment on categories that diferent peopel ave preferences with no objective. This part has essnetially a distributing matching/KMeans component; find the style furthest from competitors that matches hte largest uaidnece.

A second reason is that as $K \to \infty$, the equality condition is more stringent. For example..g, Claude and GTP might be similar on math, but on ultra hard geometry, gPT might be better. That is, our implicit K might als oincrease

A Model of Consumer Welfare TODO: A model of consumer welfare with style included

5.3 Role of Evaluation

How does evaluation impact monoculture? It centralizes reward into a zero-sum game though it is unclear what could happen against a fixed comeptitor. Optimizing for winrate smoothes out utility and causes there to be equilibira, avoiding degenerate situations where firms will invest infinitely, then nothing, then breakdown. That is, continuous feedback allows there to be reaonsable equlbra

How do specific modifications to the ranking function make a difference Optimizing for win rate induces stability, but firms optimizing ditonctnuously for rank is poteneitally destabilizing, and could have interesting implications.

Encouraging Diveristy Counterinutiively, hetergeonous F could be caused by non-public data (more private info or private models). From an eval perspectiev: 1) multiple leaderoards (Which we already do), 2) many categories. Changes to the pairwise function has little impact, though small changes to the battles could have an impact

6 Experiments

6.1 Motivating Evidence

6.2 Synthetic Experiments

Identifiability and Equilibria

Similarity and Equilibria

6.3 Identification with Real-World Data

7 Related Work

7.1 Economics and Computation

Lotteries and Contests

Leaderboards and Ranking

Coordination Games

7.2 Language Models

7.3 Economics and Markets

Collusion

Monoculture

References

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Hartman, P. and Stampacchia, G. (1966). On some non-linear elliptic differential-functional equations.

A Proofs of Equilibria Results

A.1 Proof of Theorem 2

Theorem 2 (Equilibrium Existence)**.** *Let F_i be a prodduction function and c the cost. Then there exists an equilibrium to the game if*

$$-\nabla^2 g_{i,k}(x) \succeq \sup_z \frac{\sigma''(z)}{\sigma'(z)} \cdot \nabla g_{i,k}(x) \nabla g_{i,k}(x)^\top \qquad \forall x, \ \forall i, k.$$

Proof. We prove this in three parts. We first show that there exists some maximizer for each function. We show this by the fact that the cost function, $c(\|\mathbf{x}_i\|)$, is unbounded, while the rewards are bounded due to the boundedness of σ. We next show that there exists some critical point which is a mutual best response (that is, a candidate equilibrium). Here, $\mathcal{X}$ is a convex bounded set, as any $c(\|\mathbf{x}_i\|) > 1$ leads to negative reward, and so we can restrict our attention to $\|\mathbf{x}_i\| \leq c^*$, for $c^* = c^{-1}(1)$. Then by prior work (Hartman and Stampacchia, 1966), the function $\Phi(x^*) = \langle \nabla_{x_1} U_1(x), \ldots, \nabla_{x_N} U_N(x) \rangle$ has a point so

$$\langle \Phi(x^*, x - x^* \rangle \leq 0 \forall x.$$

This exists because the space $\mathcal{X}$ is a compact convex set. Therefore, this represents a (local) critical point. We then investigate the Hessian of U to identify when a local critical point is globally optimal, resulting in the condition

$$-\nabla^2 g_{i,k}(x) \succeq \sup_z \frac{\sigma''(z)}{\sigma'(z)} \cdot \nabla g_{i,k}(x) \nabla g_{i,k}(x)^\top \qquad \forall x, \ \forall i, k.$$

□

A.2 Proof of Theorem 5

Theorem 5 (Homogeneity without Equality)**.** *Under (H1)–(H2) and (C), the set*

$$\mathcal{T} := \{\boldsymbol{\phi} \in U : H_{i,i'}(\boldsymbol{\phi}) = 0 \ \ \forall\, i, i'\}$$

is a smooth submanifold of U of dimension

$$\dim \mathcal{T} = Nk - (K-1)(N-1),$$

while the identity locus

$$\mathcal{I} := \{\boldsymbol{\phi} \in U : \phi_1 = \cdots = \phi_N\}$$

has dimension $\dim \mathcal{I} = k$ and satisfies $\mathcal{I} \subseteq \mathcal{T}$ by Lemma 1. Consequently,

$$\operatorname{codim}(\mathcal{I} \text{ in } \mathcal{T}) = (N-1)(k-K+1),$$

which is strictly positive whenever $k \geq K$. In this regime $\mathcal{I}$ has Lebesgue measure zero within $\mathcal{T}$: a generic point of $\mathcal{T}$ satisfies $H_{i,i'} = 0$ for every pair while $F_{\phi_i} \neq F_{\phi_{i'}}$ for some pair i, i'.

Proof. Fix firm 1 as a reference and define, for $i = 2, \ldots, N$, the normal-component map

$$g_i(\boldsymbol{\phi}) := \left(I_K - \frac{\theta_1(\boldsymbol{\phi})\, \theta_1(\boldsymbol{\phi})^\top}{\|\theta_1(\boldsymbol{\phi})\|^2} \right) \theta_i(\boldsymbol{\phi}).$$

By (H1)–(H2), θ_1 and θ_i are smooth and $\theta_1 \neq 0$ on U, so g_i is a smooth map on U taking values in the $(K-1)$-dimensional subspace $\theta_1(\boldsymbol{\phi})^\perp \subset \mathbb{R}^K$. By construction $g_i(\boldsymbol{\phi}) = 0$ if and only if $\theta_i(\boldsymbol{\phi}) \parallel \theta_1(\boldsymbol{\phi})$. Because "parallel to θ_1" is a transitive relation among nonzero vectors, $g_2(\boldsymbol{\phi}) = \cdots = g_N(\boldsymbol{\phi}) = 0$ holds if and only if $\theta_i(\boldsymbol{\phi}) \parallel \theta_{i'}(\boldsymbol{\phi})$ for every pair i, i', i.e., if and only if $\boldsymbol{\phi} \in \mathcal{T}$. Stack these into a single section $g := (g_2, \ldots, g_N)$ of the rank-$(K-1)(N-1)$ vector bundle $\bigoplus_{i=2}^N \theta_1(\boldsymbol{\phi})^\perp$ over U, so that $\mathcal{T} = g^{-1}(0)$.

We now check that g is transverse to the zero section, i.e., that $Dg(\boldsymbol{\phi})$ has full row rank $(K-1)(N-1)$ at every point of $\mathcal{T}$. For each $i = 2, \ldots, N$, consider the sub-block of columns of Dg corresponding to $\partial\theta_i/\partial\phi_i$ alone, holding the equilibrium-response term aside; by the chain rule this sub-block includes the direct term $D_\phi F_{\phi_i}(x_i)$, which is surjective onto $\mathbb{R}^K$ by (C).

Composing with the rank-$(K-1)$ projection defining g_i preserves surjectivity onto the $(K-1)$-dimensional target generically, i.e., unless the equilibrium-response term happens to exactly cancel the direct term's image along the projected directions, an exceptional coincidence that occurs only on a measure-zero subset of parameter configurations by the same genericity argument used throughout. Away from that exceptional set, the sub-block $\partial g_i/\partial\phi_i$ therefore has full rank $K-1$.

Since the column sets $\{\phi_i\}_{i=2}^N$ used to certify these sub-blocks are pairwise disjoint, selecting exactly these columns yields a block-diagonal $(K-1)(N-1) \times (K-1)(N-1)$ submatrix of Dg with each diagonal block of rank $K-1$, regardless of the values of any other entries of Dg, including cross-derivatives such as $\partial g_i/\partial\phi_1$ or $\partial g_i/\partial\phi_j$ for $j \neq i$ arising from the joint equilibrium coupling. This submatrix has full rank $(K-1)(N-1)$, certifying that Dg itself has full row rank $(K-1)(N-1)$, i.e., that g is transverse to the zero section. By the preimage theorem for sections of vector bundles, $\mathcal{T} = g^{-1}(0)$ is a smooth submanifold of U of dimension

$$\dim \mathcal{T} = \dim U - (K-1)(N-1) = Nk - (K-1)(N-1).$$

The identity locus $\mathcal{I}$ is the diagonal embedding of $\Phi \cong \mathbb{R}^k$ into Φ^N, hence a smooth submanifold of dimension k, and $\mathcal{I} \subseteq \mathcal{T}$ by Lemma 1. The codimension of $\mathcal{I}$ within $\mathcal{T}$ is therefore

$$\dim \mathcal{T} - \dim \mathcal{I} = Nk - (K-1)(N-1) - k = (N-1)k - (K-1)(N-1) = (N-1)(k-K+1),$$

which is strictly positive whenever $k \geq K$, so $\mathcal{I}$ is a strict, measure-zero submanifold of $\mathcal{T}$ in this regime. □

A.3 Proof of Lemma 8

Lemma 8 (Similarity in Production Functions)**.** *Suppose the following hold. First* $\sup_x \|F_i(\mathbf{x}) - F_{i'}(\mathbf{x})\| \leq \epsilon$. *Second, the gradient* $\Phi(\mathbf{x}) = \langle \nabla F_1, \nabla F_2, \ldots, \nabla F_N \rangle$ *is strongly monotonic by* μ*;*

$$(\Phi(\mathbf{x}) - \Phi(\mathbf{x}')) \cdot (\mathbf{x} - \mathbf{x}') \leq -\mu \|\mathbf{x} - \mathbf{x}'\|^2.$$

Third, F *is* L*-Lipschitz, and* $\|\nabla F_i(\mathbf{x}) - \nabla F_j(\mathbf{x})\|_\infty \leq \epsilon$*. Then*

$$\|F_i(\mathbf{x}_i^*) - F_j(x_j^*)\|_\infty \leq \epsilon(1 + L/\mu),$$

and

$$H_{i,j} \geq 1 - 2\epsilon(1 + C/\mu)/m,$$

where m *is a lower bound on* $\|\theta_i\|$*.*

Proof. We first note that

$$\|F_i(x_i^*) - F_j(x_j^*)\| \leq \|F_i(x_i^*) - F_j(x_i^*)\| + \| \leq \|F_j(x_i^*) - F_j(x_j^*)\|.$$

The first term is bounded by ϵ, while the second term is bounded by $L\|x_i^* - x_j^*\|$. We next note that at equilibrium $\Phi_i(\mathbf{x}_i^*) = \Phi_j(\mathbf{x}_j^*) = 0$ Next, by strong monotonicity,

$$\Phi_i(x_i^*) - \Phi_i(x_j^*) \cdot (x_i^* - x_j^*) \leq -\mu \|x_i^* - x_j^*\|^2.$$

We can then bound

$$\|x_i^* - x_j^*\|^2 \leq \frac{\|\Phi_i(x_j^*)\|}{\mu}.$$

We then note that $\|\Phi_i(x_j^*)\| = \|\Phi_i(x_j^*) - \Phi_j(x_j^*)\|$ only depends on F and ∇F, both of which are ϵ-similar, and so $\|\Phi_i(x_j^*)\| \leq C\epsilon$. Therefore, we have that

$$\|F_i(x_i^*) - F_j(x_j^*)\| \leq \epsilon(1 + LC/\mu) = O(\epsilon).$$

By properties of cosine similarity this gives us that

$$H_{i,j} \geq 1 - O(\epsilon).$$

□